

Certifying Robustness Transfer Between Neural Networks:

Bounds on Adversarial Confidence Margins,
a Counterexample, and a Tight Modified Formulation

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1 Abstract

We study the relationship between the adversarial robustness margins of two classifiers N_1 and N_2 , formally defined via Mixed-Integer Programming (MIP) within the VHA-GaR [1] framework. Three quantities are central: $\delta_1(N_1)$, $\delta_1(N_2)$ (the *standard-mode* MIP values — the maximum confidence each network achieves on inputs that can still be fooled by a bounded perturbation), and δ_{diff}^* (the *transfer MIP* value — the maximum gap in confidence between N_2 and N_1 at points where N_1 is robust but N_2 can be fooled).

We make four contributions:

1. We formalize all three quantities as optimization programs (Section 3).
2. We prove the *correct* inequality direction: $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$ (Theorem 4.1, Section 4).
3. We exhibit an explicit counterexample with two simple linear networks showing that the reverse inequality, $\delta_1(N_1) + \delta_{\text{diff}}^* \geq \delta_1(N_2)$, is *false* in general (Section 5).
4. We propose a *minimal modification* to the transfer MIP: replacing the N_1 -robustness constraint with an N_1 -foolability constraint. We prove that the modified quantity $\hat{\delta}_{\text{diff}}$ satisfies $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$ (Theorem 7.1, Section 7), where $\hat{\delta}_1(N_2)$ is the standard-mode value restricted to inputs on which *both* networks are simultaneously foolable.

The modified formulation admits a tight inequality, is practically computable as a MIP, and yields a new interpretation: it measures the maximum additional confidence N_2 achieves over N_1 at the most *jointly vulnerable* inputs.

2 Introduction

A central question in deep-learning safety is: *how does adversarial robustness change when a model is fine-tuned, distilled, or retrained?* Concretely, given a source model N_1 and a target model N_2 (both mapping an input $x \in [0, 1]^n$ to class logits), does improved accuracy or architecture of N_2 imply improved adversarial robustness? If not, by how much can N_2 's robustness exceed N_1 's?

The VHAGaR framework [1] answers this question rigorously via MIP. It defines:

- $\delta_1(N)$: the *hardest* correctly classified input that can still be fooled by a small perturbation (high confidence yet still fragile).
- $\delta_{\text{diff}}^*(N_1, N_2)$: the maximum *gap* in confidence between N_2 and N_1 at points where N_1 is already robust but N_2 can be attacked — a "transfer" quantity.

A natural conjecture is that $\delta_1(N_1) + \delta_{\text{diff}}^* \geq \delta_1(N_2)$: that is, the standard-mode value of N_2 cannot exceed the sum of the standard-mode value of N_1 and the transfer gap. We show this is *incorrect* in general. We then identify the precise structural reason for the failure and propose a minimal modification that restores the desired inequality.

Notation. Scalars are lowercase, networks are uppercase. We fix two output classes: $c_{\text{tag}} = 1$ (the “correct” class) and $c_{\text{target}} = 2$ (the “adversarial” class). All optimization problems are stated as linear programs (LP) or mixed-integer programs (MIP); feasibility is assumed unless stated.

3 Formal Definitions

Setup.

- Input domain: $\mathcal{X} = [0, 1]^n$.
- Perturbation set: $\mathcal{E} = [-p, p]^n$ for a fixed radius $p > 0$.
- Small constants: attack margin $\eta > 0$, transition margin $\varepsilon_0 > 0$ (both typically 10^{-3}).

Definition 3.1 (Confidence margin). For a network $N : \mathcal{X} \rightarrow \mathbb{R}^2$,

$$\text{conf}_N(x) := \text{logit}_N(x)[c_{\text{tag}}] - \text{logit}_N(x)[c_{\text{target}}].$$

$\text{conf}_N(x) > 0$ means N predicts class c_{tag} at x .

Definition 3.2 (N -foolability). $x \in \mathcal{X}$ is N -foolable if

$$\exists \varepsilon \in \mathcal{E} \text{ with } x + \varepsilon \in \mathcal{X} \text{ and } \text{conf}_N(x + \varepsilon) \leq -\eta.$$

Write $\mathcal{F}_N$ for the set of all N -foolable inputs in $\mathcal{X}$.

Definition 3.3 (Standard-mode value).

$$\delta_1(N_i) := \sup \{ \text{conf}_{N_i}(x) \mid x \in \mathcal{F}_i, \text{conf}_{N_i}(x) \geq 0 \}.$$

In words: the maximum confidence N_i attains on any input that (i) is correctly classified, yet (ii) can be misclassified by some perturbation $\varepsilon \in \mathcal{E}$.

Remark 3.1. $\delta_1(N_i)$ is computed by a MIP: maximize $\text{conf}_{N_i}(x)$ subject to the encoding of $\text{conf}_{N_i}(x + \varepsilon) \leq -\eta$ over the joint variables (x, ε) .

Definition 3.4 (Transfer MIP — original).

$$\delta_{\text{diff}}^* := \sup \left\{ \begin{array}{l} \text{conf}_{N_2}(x) - \text{conf}_{N_1}(x) \mid \\ \text{(C1) } \text{conf}_{N_1}(x) \geq \delta_1(N_1) + \varepsilon_0, \\ \text{(C2) } \text{conf}_{N_2}(x) \geq \text{conf}_{N_1}(x), \\ \text{(C3) } x \in \mathcal{F}_2 \end{array} \right\}. \quad (1)$$

Constraint C1 ensures x is N_1 -robust (it lies strictly above the highest N_1 -fragile confidence level). C2 ensures N_2 is at least as confident as N_1 at x . C3 ensures N_2 can be fooled at x .

4 The Correct Inequality

Theorem 4.1 (Lower bound on $\delta_1(N_2)$). *If the transfer MIP (1) is feasible, then*

$$\boxed{\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*}.$$

Proof. Fix $\epsilon > 0$ and let x^* be a nearly-optimal feasible point of (1), with ε^* the corresponding attack perturbation (from C3). By feasibility:

$$\text{conf}_{N_1}(x^*) \geq \delta_1(N_1) + \varepsilon_0, \quad (\text{C1})$$

$$\text{conf}_{N_2}(x^*) \geq \text{conf}_{N_1}(x^*), \quad (\text{C2})$$

$$x^* \in \mathcal{F}_2 \text{ via } \varepsilon^*, \quad (\text{C3})$$

$$\text{conf}_{N_2}(x^*) - \text{conf}_{N_1}(x^*) \geq \delta_{\text{diff}}^* - \epsilon. \quad (\text{obj})$$

Step 1 — lower-bound $\text{conf}_{N_2}(x^*)$:

$$\begin{aligned} \text{conf}_{N_2}(x^*) &= \text{conf}_{N_1}(x^*) + (\text{conf}_{N_2}(x^*) - \text{conf}_{N_1}(x^*)) \\ &\geq (\delta_1(N_1) + \varepsilon_0) + (\delta_{\text{diff}}^* - \epsilon) && (\text{C1 and obj}) \\ &\geq \delta_1(N_1) + \delta_{\text{diff}}^* - \epsilon. \end{aligned}$$

Step 2 — x^* is feasible for the standard-mode MIP on N_2 : By C2 and C1, $\text{conf}_{N_2}(x^*) \geq \text{conf}_{N_1}(x^*) \geq \delta_1(N_1) + \varepsilon_0 > 0$, so N_2 *correctly* classifies x^* . Combined with C3 ($x^* \in \mathcal{F}_2$), the pair (x^*, ε^*) is feasible for the standard-mode MIP on N_2 . Hence:

$$\delta_1(N_2) \geq \text{conf}_{N_2}(x^*) \geq \delta_1(N_1) + \delta_{\text{diff}}^* - \epsilon.$$

Since $\epsilon > 0$ is arbitrary, $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$. $\square$

Remark 4.1 (Gap quantified). Let x^* achieve the supremum in (1). The proof shows

$$\delta_1(N_2) - (\delta_1(N_1) + \delta_{\text{diff}}^*) \geq \text{conf}_{N_1}(x^*) - \delta_1(N_1) \geq \varepsilon_0 > 0.$$

The gap is *always positive*: the transfer MIP forces C1, guaranteeing $\text{conf}_{N_1}(x^*) > \delta_1(N_1)$. This surplus $\text{conf}_{N_1}(x^*) - \delta_1(N_1)$ is the precise “slack” that makes Theorem 4.1 strict and the reverse inequality false.

5 The Reverse Inequality Fails: a Counterexample

5.1 Network Definitions

We exhibit two explicit linear networks over $\mathcal{X} = [0, 1]^2$, with $p = 0.1$, $\eta = \varepsilon_0 = 10^{-3}$.

Network	Logits	Confidence margin
N_1	$[x_1, x_2]$	$\text{conf}_{N_1}(x) = x_1 - x_2$
N_2	$[2x_1, 5x_2]$	$\text{conf}_{N_2}(x) = 2x_1 - 5x_2$

Both are two-layer sequential networks: an identity first layer, an inactive ReLU (since $x_1, x_2 \geq 0$), and a diagonal second layer ($\text{diag}(1, 1)$ for N_1 , $\text{diag}(2, 5)$ for N_2).

5.2 Foolability Conditions

N_1 . N_1 is fooled at $x + \varepsilon$ iff $(x_2 + \varepsilon_2) - (x_1 + \varepsilon_1) \geq \eta$, i.e., $\varepsilon_2 - \varepsilon_1 \geq (x_1 - x_2) + \eta$. The maximum of $\varepsilon_2 - \varepsilon_1$ over $\mathcal{E}$ is $2p = 0.2$, so:

$$x \in \mathcal{F}_1 \iff x_1 - x_2 \leq 2p - \eta = 0.199.$$

N_2 . N_2 is fooled at $x + \varepsilon$ iff $5(x_2 + \varepsilon_2) - 2(x_1 + \varepsilon_1) \geq \eta$, i.e., $5\varepsilon_2 - 2\varepsilon_1 \geq (2x_1 - 5x_2) + \eta$. The maximum of $5\varepsilon_2 - 2\varepsilon_1$ over $\mathcal{E}$ is $5p + 2p = 0.7$, so:

$$x \in \mathcal{F}_2 \iff 2x_1 - 5x_2 \leq 5p + 2p - \eta = 0.699.$$

5.3 Computing $\delta_1(N_1)$, $\delta_1(N_2)$, and δ_{diff}^*

$\delta_1(N_1)$. Maximize $x_1 - x_2$ subject to $x_1 - x_2 \leq 0.199$ and $x \in [0, 1]^2$:

$$\delta_1(N_1) = 0.199.$$

Achieved at $x^{(1)} = (1, 0.801)$ (or equivalently at any x on the boundary $x_1 - x_2 = 0.199$).

$\delta_1(N_2)$. Maximize $2x_1 - 5x_2$ subject to $2x_1 - 5x_2 \leq 0.699$ and $x \in [0, 1]^2$:

$$\delta_1(N_2) = 0.699.$$

Achieved at $x^{(2)} = (1, 0.2602)$ with attack $\varepsilon^* = (-0.1, +0.1)$.

δ_{diff}^* . The objective $\text{conf}_{N_2}(x) - \text{conf}_{N_1}(x) = (2x_1 - 5x_2) - (x_1 - x_2) = x_1 - 4x_2$. Constraints:

$$\begin{aligned} \text{C1 : } & x_1 - x_2 \geq 0.200, \\ \text{C2 : } & x_1 - 4x_2 \geq 0, \\ \text{C3 : } & 2x_1 - 5x_2 \leq 0.699. \end{aligned}$$

Setting $x_2 = 0$ (minimises the $-4x_2$ penalty), C1 gives $x_1 \geq 0.200$ and C3 gives $x_1 \leq 0.3495$. Maximising x_1 yields:

$$x^* = (0.3495, 0), \quad \delta_{\text{diff}}^* = x_1^* - 4 \cdot 0 = \mathbf{0.3495}.$$

Observe: $\text{conf}_{N_1}(x^*) = 0.3495 \gg \delta_1(N_1) = 0.199$.

5.4 The Counterexample

Quantity	Value
$\delta_1(N_1)$	0.199
δ_{diff}^*	0.3495
$\delta_1(N_1) + \delta_{\text{diff}}^*$	0.5485
$\delta_1(N_2)$	0.699
Gap: $\delta_1(N_2) - (\delta_1(N_1) + \delta_{\text{diff}}^*)$	0.1505

$$\delta_1(N_1) + \delta_{\text{diff}}^* = 0.5485 < 0.699 = \delta_1(N_2).$$

The gap equals exactly $\text{conf}_{N_1}(x^*) - \delta_1(N_1) = 0.3495 - 0.199 = 0.1505$, consistent with Remark 4.1.

6 Root-Cause Analysis

Theorem 4.1 and the counterexample together show that the formula $\delta_1(N_1) + \delta_{\text{diff}}^*$ is a *strict lower bound* on $\delta_1(N_2)$, with gap

$$\text{gap} = \underbrace{\text{conf}_{N_1}(x^*)}_{\geq \delta_1(N_1) + \varepsilon_0 \text{ (C1)}} - \delta_1(N_1) \geq \varepsilon_0 > 0.$$

The structural reason is as follows. The transfer MIP searches for an x^* that simultaneously satisfies C1 (high N_1 -confidence) and C3 (N_2 -foolable). Because N_1 and N_2 have *different* decision boundaries, the N_2 -foolable region can contain points with

N_1 -confidence far above $\delta_1(N_1)$. At such an x^* , the quantity $\text{conf}_{N_2}(x^*) = \text{conf}_{N_1}(x^*) + \delta_{\text{diff}}^*$ is large, but the formula $\delta_1(N_1) + \delta_{\text{diff}}^*$ loses the surplus $\text{conf}_{N_1}(x^*) - \delta_1(N_1)$.

Equivalently: the transfer MIP's feasible set $\{x : \text{C1} + \text{C3}\}$ is a *proper subset* of the standard-mode feasible set for N_2 ($\{x \in \mathcal{F}_2\}$), so

$$\delta_1(N_2) = \sup_{x \in \mathcal{F}_2} \text{conf}_{N_2}(x) \geq \sup_{x: \text{C1} + \text{C3}} \text{conf}_{N_2}(x) \geq \sup_{x: \text{C1} + \text{C2} + \text{C3}} [\text{conf}_{N_1}(x) + \delta_{\text{diff}}^*] \geq \delta_1(N_1) + \delta_{\text{diff}}^*.$$

The first inequality can be strict (and is, in our counterexample) because $\mathcal{F}_2$ contains points with high $\text{conf}_{N_2}(\cdot)$ that *violate* C1 (they are not N_1 -robust).

7 A Modified Formulation with the Desired Inequality

7.1 Motivation

Remark 4.1 reveals that the gap is driven by C1 (the N_1 -robustness constraint) forcing $\text{conf}_{N_1}(x^*) > \delta_1(N_1)$. To close this gap, we replace C1 with its *logical opposite*: instead of requiring N_1 to be *robust* at x , we require N_1 to be *foolable* at x . This simultaneously restricts the standard-mode problem for N_2 to the same joint-foolability region.

7.2 Modified Definitions

Definition 7.1 (Joint-foolable set).

$$\mathcal{F}_{1 \wedge 2} := \mathcal{F}_1 \cap \mathcal{F}_2 = \{x \in \mathcal{X} : x \text{ is both } N_1\text{-foolable and } N_2\text{-foolable}\}.$$

Definition 7.2 (Modified standard-mode value).

$$\hat{\delta}_1(N_2) := \sup \{ \text{conf}_{N_2}(x) \mid x \in \mathcal{F}_{1 \wedge 2}, \text{conf}_{N_2}(x) \geq 0 \}.$$

This is the maximum confidence N_2 achieves at inputs that are simultaneously foolable by *both* networks.

Remark 7.1. By definition, $\hat{\delta}_1(N_2) \leq \delta_1(N_2)$: restricting to $\mathcal{F}_{1 \wedge 2} \subseteq \mathcal{F}_2$ can only decrease the supremum. The gap $\delta_1(N_2) - \hat{\delta}_1(N_2)$ measures the advantage N_2 has at points where it is foolable but N_1 is not (the “exclusive vulnerability” of N_2).

Definition 7.3 (Modified transfer MIP).

$$\begin{aligned} \hat{\delta}_{\text{diff}} := \sup \Big\{ & \text{conf}_{N_2}(x) - \text{conf}_{N_1}(x) \mid \\ & (\widehat{\text{C1}}) \ x \in \mathcal{F}_1, \\ & (\text{C3}) \ x \in \mathcal{F}_2 \Big\}. \end{aligned} \tag{2}$$

The key change: constraint C1 is replaced by $\widehat{\text{C1}}$, which requires x to be *N1-foolable* (instead of N1-robust). Constraint C2 is dropped because, over $\mathcal{F}_{1\wedge 2}$, the objective $\text{conf}_{N_2}(x) - \text{conf}_{N_1}(x)$ is unconstrained in sign; including it would restrict the feasible set without being necessary for the theorem below.

MIP encoding. $\widehat{\text{C1}}$ adds a second perturbation variable $\varepsilon^{(1)} \in \mathcal{E}$ with the foolability constraint $\text{conf}_{N_1}(x + \varepsilon^{(1)}) \leq -\eta$, encoded exactly as in the standard-mode MIP for N_1 . C3 is encoded as before. The resulting program has $O(|N_1| + |N_2|)$ binary variables and is solvable by Gurobi or any MILP solver.

7.3 Main Result

Theorem 7.1 (Upper bound on the joint standard-mode value).

$$\boxed{\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2).}$$

Proof. Fix $\epsilon > 0$ and let x^{**} be a nearly-optimal feasible point of $\hat{\delta}_1(N_2)$ (Definition 7.2), i.e.,

$$x^{**} \in \mathcal{F}_{1\wedge 2}, \quad \text{conf}_{N_2}(x^{**}) \geq 0, \quad \text{conf}_{N_2}(x^{**}) \geq \hat{\delta}_1(N_2) - \epsilon.$$

Step 1 — x^{} satisfies $\delta_1(N_1)$'s definition:** Since $x^{**} \in \mathcal{F}_1$, by Definition 3.3,

$$\text{conf}_{N_1}(x^{**}) \leq \delta_1(N_1).$$

Step 2 — x^{} is feasible for (2):** $x^{**} \in \mathcal{F}_1$ satisfies $\widehat{\text{C1}}$, and $x^{**} \in \mathcal{F}_2$ satisfies C3. Hence x^{**} is feasible for the modified transfer MIP (2). Therefore:

$$\hat{\delta}_{\text{diff}} \geq \text{conf}_{N_2}(x^{**}) - \text{conf}_{N_1}(x^{**}).$$

Step 3 — combine:

$$\begin{aligned} \delta_1(N_1) + \hat{\delta}_{\text{diff}} &\geq \delta_1(N_1) + \text{conf}_{N_2}(x^{**}) - \text{conf}_{N_1}(x^{**}) \\ &\geq \text{conf}_{N_1}(x^{**}) + \text{conf}_{N_2}(x^{**}) - \text{conf}_{N_1}(x^{**}) \quad (\text{Step 1: } \delta_1(N_1) \geq \text{conf}_{N_1}(x^{**})) \\ &= \text{conf}_{N_2}(x^{**}) \\ &\geq \hat{\delta}_1(N_2) - \epsilon. \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$. □

Corollary 7.2 (Sandwich bounds). *If both the original and modified transfer MIPs are feasible:*

$$\delta_1(N_1) + \delta_{\text{diff}}^* \leq \delta_1(N_2), \quad \delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2).$$

Remark 7.2 (Relationship between δ_{diff}^* and $\hat{\delta}_{\text{diff}}$). The two transfer quantities are generally incomparable:

- The original δ_{diff}^* searches over N_1 -robust points (C1 requires $\text{conf}_{N_1}(x) \geq \delta_1(N_1) + \varepsilon_0$), finding large gaps where N_2 fails but N_1 does not.
- The modified $\hat{\delta}_{\text{diff}}$ searches over N_1 -foolable points, i.e., the intersection $\mathcal{F}_{1 \wedge 2}$. Because $\mathcal{F}_{1 \wedge 2}$ is disjoint from the original C1 feasible set, the two MIPs explore *complementary regions* of $\mathcal{X}$.

In particular $\hat{\delta}_{\text{diff}}$ can be negative (if, in $\mathcal{F}_{1 \wedge 2}$, N_1 always has higher confidence than N_2), while the original δ_{diff}^* is non-negative by C2.

7.4 Interpretation

Original δ_{diff}^* — transfer attack perspective. The original formulation asks: “By how much can N_2 ’s confidence *exceed* N_1 ’s, at a point where N_1 is provably robust?” This is the *hardest* transfer scenario: N_1 is safe, N_2 is not. Theorem 4.1 shows this provides a *lower bound* on $\delta_1(N_2) - \delta_1(N_1)$.

Modified $\hat{\delta}_{\text{diff}}$ — joint vulnerability perspective. The modified formulation asks: “At points that are simultaneously the hardest for *both* networks, how much more confident is N_2 than N_1 ?” Theorem 7.1 shows this provides an *upper bound* on $\hat{\delta}_1(N_2) - \delta_1(N_1)$.

Together, the two quantities bracket the robustness difference from both sides, over their respective domains.

8 Numerical Verification

8.1 Computing $\hat{\delta}_{\text{diff}}$ for the Counterexample Networks

Recall N_1, N_2 from Section 5, with $p = 0.1$, $\eta = \varepsilon_0 = 10^{-3}$.

The modified transfer MIP (2) maximizes $\text{conf}_{N_2}(x) - \text{conf}_{N_1}(x) = x_1 - 4x_2$ subject to:

$$\begin{aligned} \widehat{\text{C1}} : \quad & x_1 - x_2 \leq 0.199 \quad (N_1\text{-foolable}), \\ \text{C3} : \quad & 2x_1 - 5x_2 \leq 0.699 \quad (N_2\text{-foolable}), \\ & x \in [0, 1]^2. \end{aligned}$$

Setting $x_2 = 0$: $x_1 \leq \min(0.199, 0.3495) = 0.199$. Maximising x_1 gives $x_1^{**} = 0.199$, so:

$$\hat{\delta}_{\text{diff}} = 0.199 - 4 \cdot 0 = \mathbf{0.199}.$$

8.2 Computing $\hat{\delta}_1(N_2)$

The joint standard-mode MIP maximizes $2x_1 - 5x_2$ subject to both N_1 -foolable and N_2 -foolable constraints:

$$x_1 - x_2 \leq 0.199, \quad 2x_1 - 5x_2 \leq 0.699, \quad x \in [0, 1]^2.$$

At $x_2 = 0$: $x_1 \leq 0.199$. Maximum of $2x_1$ is $2(0.199)$:

$$\hat{\delta}_1(N_2) = 2 \times 0.199 = \mathbf{0.398}.$$

8.3 Verification of Theorem 7.1

Quantity	Value
$\delta_1(N_1)$	0.199
$\hat{\delta}_{\text{diff}}$ (modified transfer MIP)	0.199
$\delta_1(N_1) + \hat{\delta}_{\text{diff}}$	0.398
$\hat{\delta}_1(N_2)$ (joint standard-mode, N_2)	0.398
$\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$?	YES (equality)
$\delta_1(N_2)$ (unrestricted, for reference)	0.699
Joint gap: $\delta_1(N_2) - \hat{\delta}_1(N_2)$	0.301

$$\delta_1(N_1) + \hat{\delta}_{\text{diff}} = 0.199 + 0.199 = 0.398 = \hat{\delta}_1(N_2). \quad \checkmark$$

For these networks, equality holds in Theorem 7.1 because the binding constraint is simultaneously $\widehat{\text{C1}}$ (N_1 -foolability) and C3 (N_2 -foolability), making x^{**} the argmax for both $\hat{\delta}_1(N_2)$ and $\hat{\delta}_{\text{diff}}$.

The “joint gap” $\delta_1(N_2) - \hat{\delta}_1(N_2) = 0.301$ reflects the portion of N_2 ’s foolable region that is *N1-robust* — precisely the regime probed by the original transfer MIP.

9 Discussion

Two complementary MIPs. The original and modified transfer MIPs are not competing; they probe complementary aspects of the two-network robustness landscape:

MIP	Feasible region	Bound direction
Original δ_{diff}^*	N_1 -robust $\cap$ N_2 -foolable	$\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$ (lower bound)
Modified $\hat{\delta}_{\text{diff}}$	N_1 -foolable $\cap$ N_2 -foolable	$\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$ (upper bound)

When are the bounds tight? The original lower bound $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$ is tight when $\text{conf}_{N_1}(x^*) = \delta_1(N_1)$ at the transfer optimum, i.e., when the most vulnerable N_2 point (while N_1 is robust) is also the hardest N_1 -fragile point. The modified upper bound $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$ is tight (equality) when the argmax of $\hat{\delta}_1(N_2)$ and $\hat{\delta}_{\text{diff}}$ coincide, as in our numerical example.

Practical implications.

- *Model comparison:* If N_2 is a distilled or retrained version of N_1 , computing δ_{diff}^* gives a certificate that $\delta_1(N_2)$ has increased by at least δ_{diff}^* . Computing $\hat{\delta}_{\text{diff}}$ gives a certificate that the improvement is bounded above by $\delta_1(N_1) + \hat{\delta}_{\text{diff}} - \delta_1(N_1) = \hat{\delta}_{\text{diff}}$ within the jointly vulnerable region.
- *Audit scenarios:* The joint-foolable region $\mathcal{F}_{1 \wedge 2}$ is the set of inputs most dangerous for a *deployed ensemble* of N_1 and N_2 . $\hat{\delta}_1(N_2)$ quantifies the worst-case overconfidence within this region.
- *Training objectives:* A model trainer who wishes to improve $\hat{\delta}_1(N_2)$ (reduce N_2 's fragility in $\mathcal{F}_{1 \wedge 2}$) can monitor $\delta_1(N_1) + \hat{\delta}_{\text{diff}}$ as a computable upper bound on this quantity.

10 Conclusion

We studied the three MIP-defined robustness quantities $\delta_1(N_1)$, $\delta_1(N_2)$, δ_{diff}^* arising in the VHAGaR neural-network verification framework. Our results are:

1. **Theorem 4.1:** $\delta_1(N_2) \geq \delta_1(N_1) + \delta_{\text{diff}}^*$ always. The original transfer MIP provides a *lower bound* on the improvement in robustness from N_1 to N_2 .
2. **Counterexample** (Section 5): the reverse inequality $\delta_1(N_1) + \delta_{\text{diff}}^* \geq \delta_1(N_2)$ is false in general; the gap equals the surplus N_1 -confidence at the transfer optimum.
3. **Theorem 7.1:** by flipping C1 from N_1 -robustness to N_1 -foolability, the modified transfer MIP quantity $\hat{\delta}_{\text{diff}}$ satisfies $\delta_1(N_1) + \hat{\delta}_{\text{diff}} \geq \hat{\delta}_1(N_2)$, providing a computable *upper bound* on robustness within the joint-foolable region.

Together, the two MIPs provide a two-sided characterization of the robustness relationship between any pair of networks. The precise bound one computes depends on whether the analyst wishes to guarantee that N_2 is *at least as robust* as $N_1 + \delta_{\text{diff}}^*$ (original), or to certify that N_2 's vulnerability in the hardest jointly- foolable regime is bounded by $\delta_1(N_1) + \hat{\delta}_{\text{diff}}$ (modified).

Future work. A natural extension is to generalise from two classes to $k > 2$ classes, to non- ℓ_∞ perturbation sets, and to establish analogous bounds in the case where N_1 and N_2 share a backbone but differ in the classification head (the distillation scenario).

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